



Stair-Climbing Robotic Dog for Oxygen Therapy Patients

An autonomous quadruped that follows a patient on supplemental oxygen, detects a staircase, and climbs it carrying the tank — freeing both hands for the railing.
Team Robodog (Project #6)

TASH 2026

MOTIVATION

Patients on long-term oxygen therapy must haul a portable concentrator everywhere — stairs are the **highest fall-risk, highest-exertion obstacle** in the home.

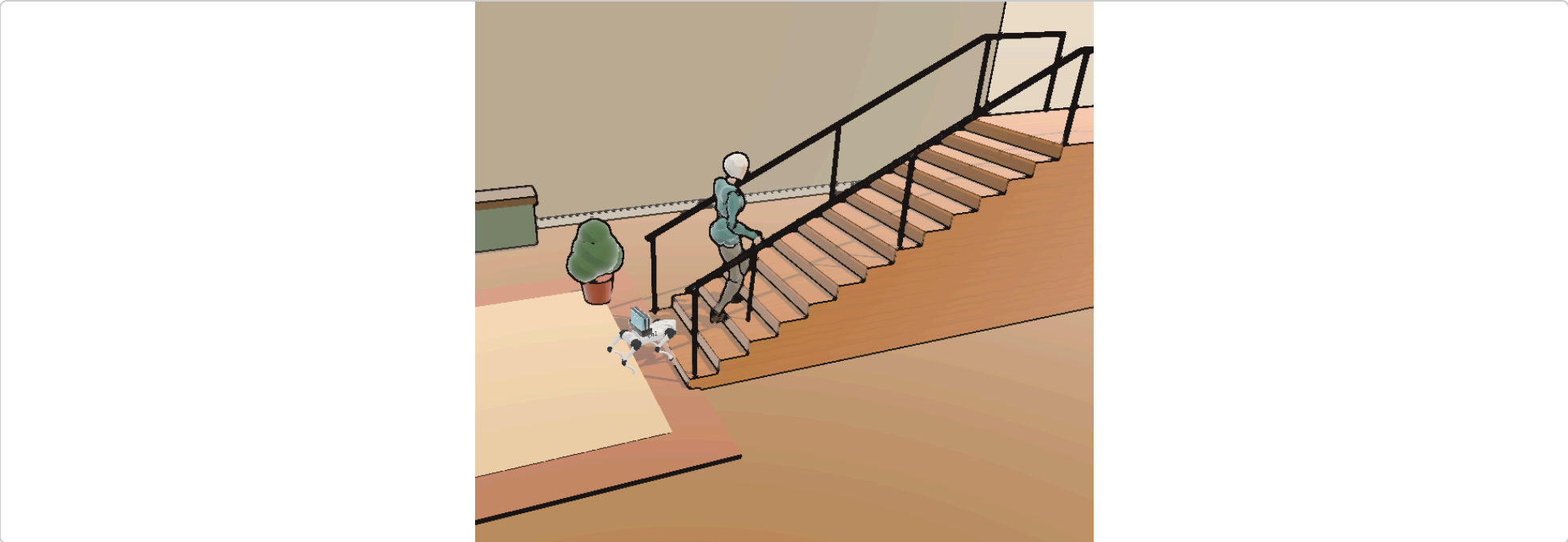


Fig. 1 — The patient still climbs unassisted today; the robot in this Isaac Sim scene is the fix we're building toward.

- Older stair-users show higher stride variability & fall risk (Startzell 2000; Hausdorff 1997) — an extra device only compounds that.
- Our robot **carries the ~2.2 kg concentrator**, follows autonomously, detects stairs, and climbs — beside the patient.
- Both hands freed for the railing; robot handles the load.
- The goal isn't to replace the patient's independence — it's to remove the one obstacle (the tank) that takes it away.

HARDWARE PLATFORM



Robot: Unimove Go2 , 6.9 kg, 12-DOF
Compute: NVIDIA Jetson Orin
Vision: Intel RealSense D435 RGB-D
LiDAR: Hesai XT16
Payload: 2.22 kg mock O₂ + cradle

Fig. 2 — Go2 with payload cradle.

WALKING POLICY — IT FOLLOWS YOU

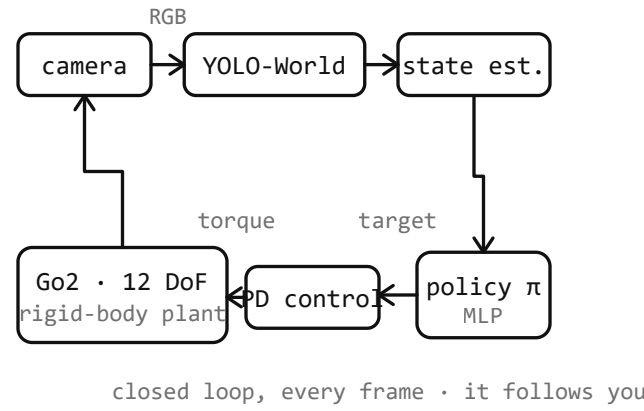


Fig. 3 — Walking-policy control loop: YOLO-World relocates the patient every frame; policy π (MLP) and a PD loop turn that into torque on the Go2.

YOLO-World re-detects the patient every frame; the **policy π** MLP turns that state into joint targets, and a PD loop converts them into torque — closing the loop so the walk policy keeps the Go2 beside the patient.

WE TRAINED IT TO CLIMB — 6,000 ITERATIONS

A blind reinforcement-learning policy learned to drive the payload upstairs on feel alone.

95.7

Mean reward
at plateau

138 mm

Riser the curriculum
settled at

≈100%

Episodes end upright,
not a fall

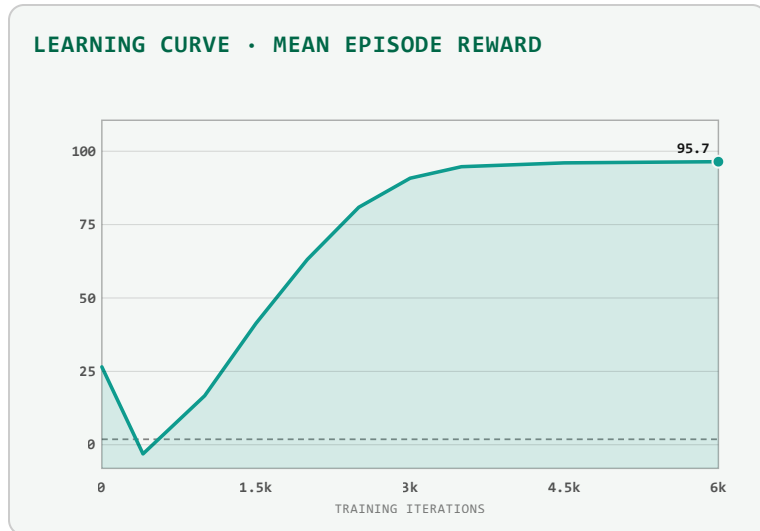


Fig. 4 — Mean episode reward climbs to 95.7 and holds over 6,000 training iterations.

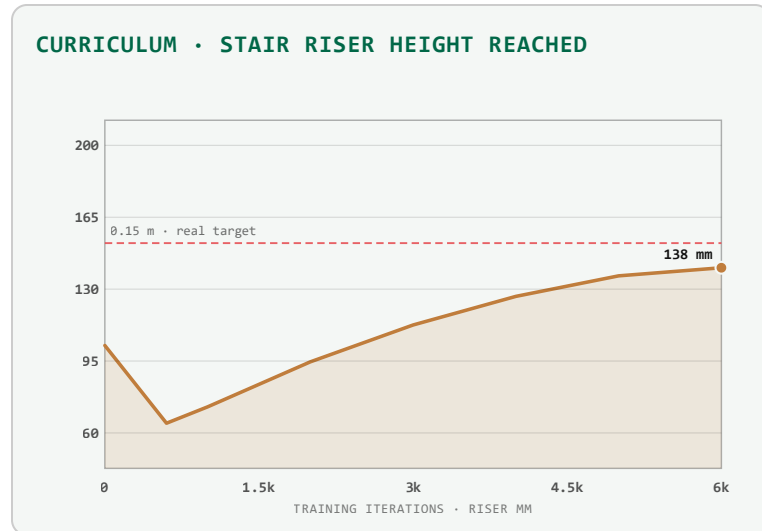


Fig. 5 — Difficulty curriculum settles at a 138 mm riser — right at the real 0.15 m target stair.

Across **6,000 training iterations** the mean episode reward climbed to **95.7** and held. A difficulty curriculum pushed the stairs steeper over time, settling at a **138 mm riser** — right at the real-world target height. Along the way the policy learned to stop falling — topples collapsed from **92% of episodes to about 3%** — and to survive longer, carrying the payload the full climb instead of tipping in the first couple of seconds. The gait it found is cautious, not graceful — it noses down and leans into each riser rather than stepping cleanly — but inside that trained envelope it ends upright almost every time, keeping the payload level.

METHODOLOGY: PERCEIVE → FOLLOW → DETECT → CLIMB

One control loop, running entirely on the robot's onboard Jetson Orin, turns raw camera and LiDAR frames into joint commands in five stages — no cloud round-trip, so the loop keeps running even if Wi-Fi drops mid-climb:

- Perceive** — RealSense D435 RGB-D + LiDAR feed two parallel vision heads: **YOLOv8-Pose** (person detection) and **YOLO-World** (open-vocab stair detection).
- Follow** — **ByteTrack** lock with 3 s occlusion hold. 3-zone PID holds **0.45 m standoff**; depth-reactive avoidance routes around furniture.
- Detect Stairs** — YOLO-World counts risers; LiDAR depth corroborates (0.6 weight blend). **≥2 risers within 1.6 m** triggers handoff. Dual-evidence gate eliminates false positives.
- Hand Off** — **50 Hz HandoffController** (WALK $\rightleftharpoons$ CLIMB → TOP-EGRESS / ABORT) swaps PGTT walker for blind-RL climb policy on motion-stall + riser trigger.
- Climb** — Proprioceptive "blind" RL policy drives legs up; at crest the robot walks clear and returns to walk policy.

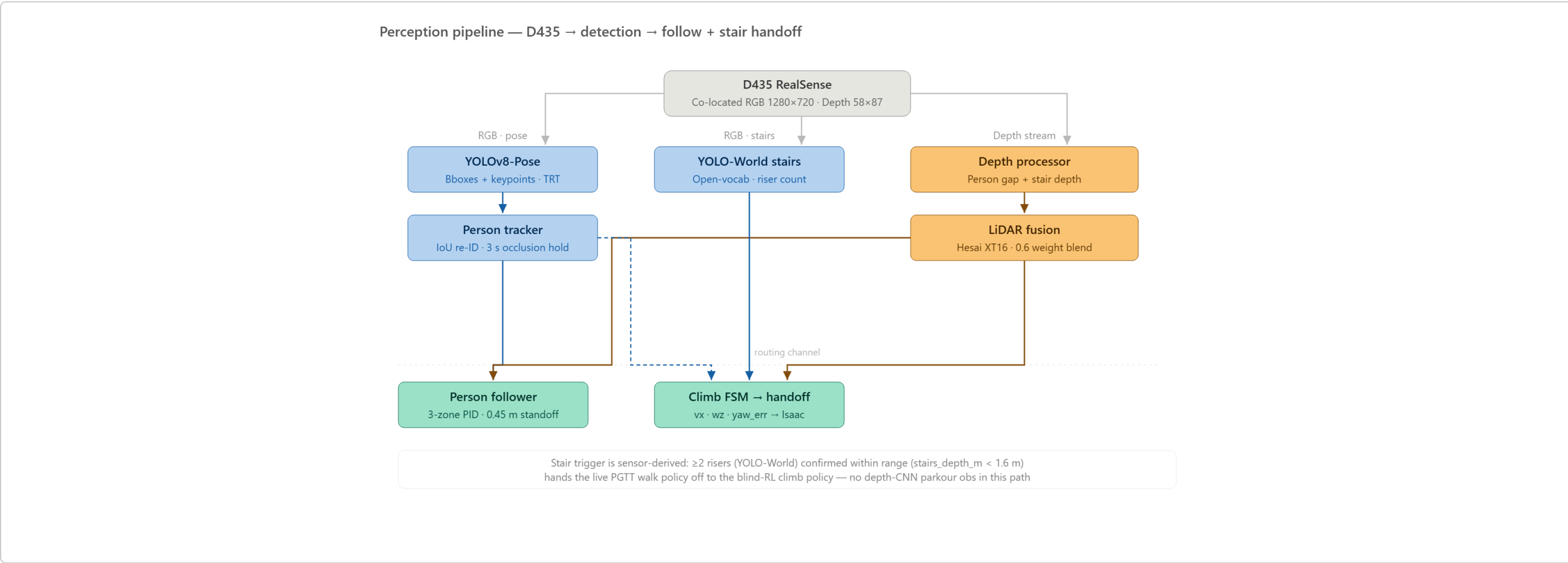


Fig. 6 — Perception pipeline: sensor inputs feed parallel vision heads; outputs merge in the Climb FSM.

The two vision heads run independently and never block each other — losing the person for a moment doesn't stall stair detection, and vice versa.

RESULTS — ISAAC SIM STAIR-HEIGHT SWEEP

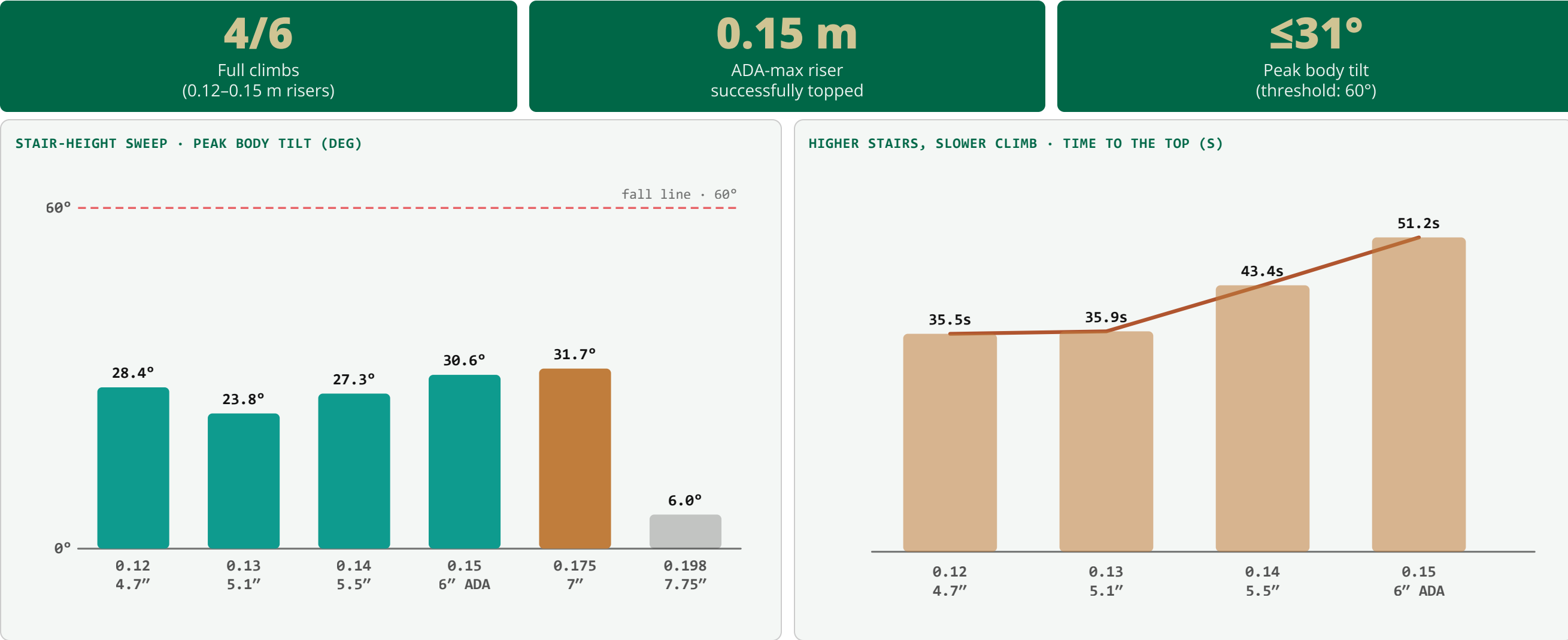


Fig. 7 — Peak body tilt at each riser height, well below the 60° fall line.

Fig. 8 — Climb time rises from 35.5 s to 51.2 s as the riser grows.

Riser	Steps (/ 14)	Peak Tilt	Outcome
0.120 m (~5")	14.0	28°	✓ Full climb
0.130 m (~5")	13.9	24°	✓ Full climb
0.140 m (~6")	13.9	27°	✓ Full climb
0.150 m (6" ADA)	14.0	31°	✓ Full climb
0.175 m (7" IBC)	4.4	32°	Slows, stalls part-way
0.198 m (7.75")	0.0	6°	Stalls at stairs

Robot topped the ADA-maximum (0.15 m) staircase end-to-end in simulation, staying upright throughout (peak tilt ≤31°, well below the 60° fall threshold).

Past code-legal risers the gait rears up against the step instead of driving through it — the same conservative "climb by feel" behavior that keeps it upright at 0.15 m becomes overly cautious once the geometry outruns what the policy was trained on.

WHY THIS MATTERS

About **1.5 million Americans** rely on long-term home oxygen therapy, and stairs are consistently the task they fear most — a fall risk that can confine a patient to one floor of their own home, cutting off bedrooms, bathrooms, or exits on other levels. This project builds a sensor-guided quadruped that walks beside the patient, carries the oxygen equipment, and climbs the stairs on its own — so the patient never has to face a staircase alone.

CONTROL STATE MACHINES

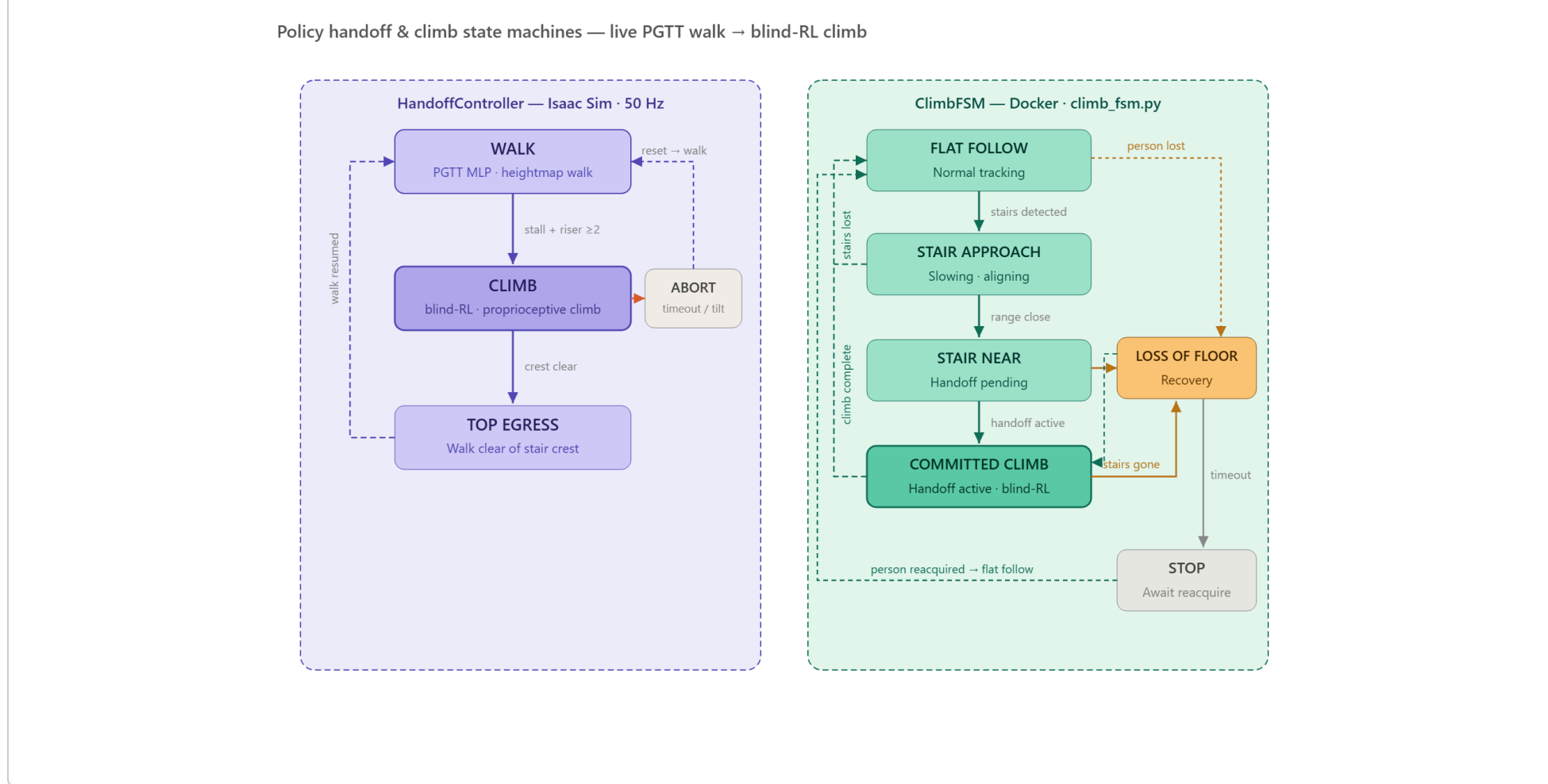


Fig. 9 — HandoffController (Isaac Sim, 50 Hz) and ClimbFSM (Docker) run concurrently via UDP.

Two state machines, one on each side of the **UDP** link, so a dropped packet or a stalled **ClimbFSM** can't leave the walker holding a stale climb command.

SIMULATION & TRAINING

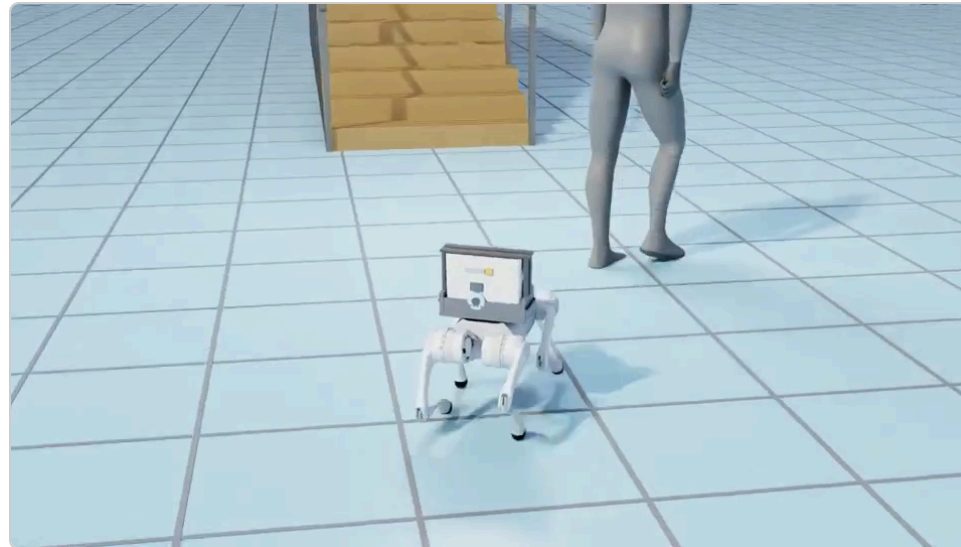
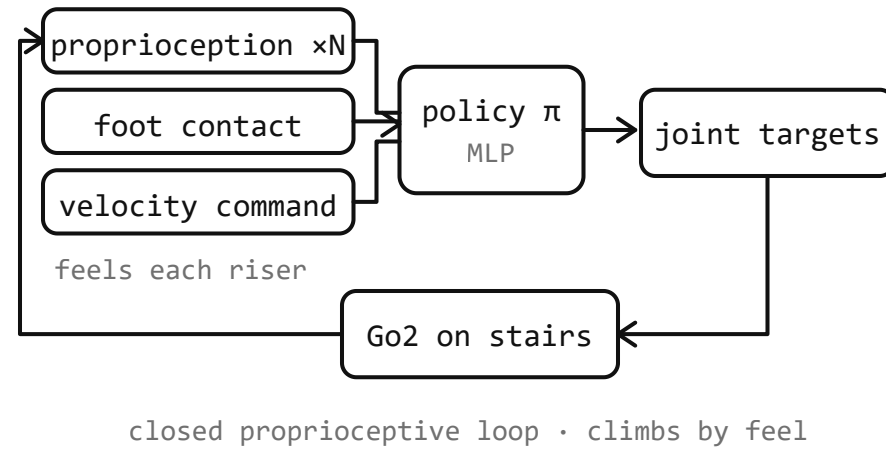


Fig. 10 — Scene view (Isaac Sim rollout): robot trails the patient toward the stairs, carrying the O₂ tank.



Fig. 11 — Live perception HUD: YOLO-World person lock, D435 depth, and LiDAR range panels.

BLIND-RL CLIMB POLICY



closed proprioceptive loop · climbs by feel

Fig. 12 — Blind-RL policy: no camera, no depth on the stairs — terrain is felt via proprioception + foot contact alone.

Going blind on the stairs was a deliberate choice, not a limitation of the cameras — vision is unreliable right at the robot's feet, where its own body occludes the D435's view of the next riser.

CONCLUSIONS

- End-to-end **follow** → **detect** → **hand-off** → **climb** stack validated in NVIDIA Isaac Sim on a shared sim/real codebase (**401 passing tests**).
- Successfully reaches the top of the **0.15 m (ADA-maximum)** staircase end-to-end in the best demo.
- Peak tilt ≤31° across all successful runs — **well below the 60° fall threshold**.
- Same policy and perception code run in sim and on the physical Go2 — a design choice meant to shorten the path from this sim result to a hardware demo.

LIMITATIONS & FUTURE WORK

- High CoG** — O₂ carrier raises center of gravity; wider stance being explored.
- Patient clearance** — no collisions with the patient in any simulated climb; stance-locking tuning on the incline is ongoing.
- Hardware climb unproven** — physical Go2 validation is the active next step.

ACKNOWLEDGEMENTS

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